Appendix: abelian groups

This appendix presents an account of the elementary theory of abelian groups which is heavily biased towards quotient groups, exact sequences and presentations, these being the main ideas from the theory which are used in the body of the book. In fact scarcely anything is said which is not strictly relevant to the material of the book, the only exceptions being a few partially worked examples. There are accounts of the general theory of abelian groups in many texts on algebra or group theory, for example in the books of Lang and Mac Lane & Birkhoff listed in the References. These books give full details of the classification theorem for finitely generated abelian groups, stated but not proved in A.27. However, every result from abelian group theory which is actually used in the book is proved in this appendix.

One piece of notation: here and in the text the value of a map f at an element x in its domain is denoted by fx rather than $f(x)$, unless there is a special need for bracketing as in $f(x+y)$.

Basic definitions

A.1. Abelian group An *abelian group* is a set A, together with a binary operation $+$ defined on A and satisfying the following four axioms:

(1) For all a, b, c in A, $a + (b + c) = (a + b) + c$.
(2) There exists $0 \in A$ such that, for all $a \in A$,

$$a + 0 = 0 + a = a.$$

(3) For each $a \in A$ there exists $b \in A$ such that

$$a + b = b + a = 0.$$

(4) For all a, b in A, $a + b = b + a$.

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It follows from these axioms that the 0 of (2) is unique and that, in (3), b is uniquely determined by a. We write $-a$ for this unique b and abbreviate $c + (-a)$ to $c - a$ for any $c \in A$.

Throughout this appendix the letters A, B, C will, unless otherwise stated, stand for arbitrary abelian groups. As is customary, we refer to a group using just the name of its underlying set.

A.2. Subgroup A *subgroup* of A is a subset of A which is also an abelian group under the same binary operation.

It follows that $B \subset A$ is a subgroup of A if and only if B is non-empty and, for all $a, b \in B$, we have $a - b \in B$.

A.3. Trivial group Any group with precisely one element is denoted 0 and called a *trivial* or *zero group*.

A.4. Let $a \in A$ and $n \in \mathbb{Z}$. By na we mean 0 if $n = 0$; $a + a + \cdots + a$ (n summands) if $n > 0$; and $(-a) + \cdots + (-a)$ ($-n$ summands) if $n < 0$. thus $(-n)a = -(na)$ for any n.

A.5. Order Let $a \in A$. If, for all $n \in \mathbb{Z}$, $n \neq 0$, we have $na \neq 0$ then a is said to have *infinite order*. Otherwise, the *order* of a is the smallest integer $n > 0$ such that $na = 0$. Thus a has order 1 if and only if $a = 0$ and, for any a, a and $-a$ have the same order, or both have infinite order. The *order* $|A|$ *of a group* A is simply the number of elements in A.

A.6. Cyclic group A is called *cyclic* if there exists $a \in A$ such that any $b \in A$ is of the form na for some $n \in \mathbb{Z}$. Such an a is called a *generator* of A. Note that $-a$ is also a generator.

Examples: (1) The integers $\mathbb{Z}$ under addition is a cyclic group with generator 1 (or, alternatively, -1), which has infinite order.

(2) The set $\{0, 1, \dots, k-1\}(k \geqslant 1)$ under addition modulo k is a cyclic group $\mathbb{Z}_k$ with generator 1 (or, in fact, any element coprime to k), which has order k. Note that $\mathbb{Z}_1 = 0$.

Note that we use the notation $\mathbb{Z}_k$ for a cyclic group of order k. Other common notations are $\mathbb{Z}/k$ or $\mathbb{Z}/k\mathbb{Z}$.

(3) For any A and $a \in A$ the subset $\{na : n \in \mathbb{Z}\}$ is a subgroup of A which is cyclic, a being a generator. It is called the subgroup *generated by* a and has order equal to the order of a (infinite order if a has infinite order).

A.7. Homomorphism A *homomorphism* f from A to B is a map $f : A \to B$ such that, for all $a_1, a_2 \in A, f(a_1 + a_2) = fa_1 + fa_2$. (Here the binary operations

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in both groups are denoted $+$.) This implies $f0 = 0$ and $f(-a) = -(fa)$ for any $a \in A$.

The *domain* of f is A and the *target* of f is B.

Examples: (1) The *trivial* homomorphism $f : A \to B$ is defined by $fa = 0$ for all $a \in A$; we write $f = 0$. (Thus '0' now has three meanings besides its customary one of 'the integer zero'.) Any homomorphism $0 \to A$ or $A \to 0$ is trivial.

(2) The identity map $1_A : A \to A$ is a homomorphism. If A is a subgroup of B then the inclusion $i : A \to B$ defined by $ia = a$ for all $a \in A$ is a homomorphism.

(3) If $f : A \to B$ and $g : B \to C$ are homomorphisms then so is the composite $g \circ f : A \to C$, defined by $(g \circ f)a = g(fa)$ for all $a \in A$.

A.8. Kernel, image, monic, epic, isomorphism Let $f : A \to B$ be a homomorphism. We define

$$\textit{Kernel} \text{ of } f = \operatorname{Ker} f = \{a \in A : fa = 0\};$$
$$\textit{Image} \text{ of } f = \operatorname{Im} f = \{b \in B : b = fa \text{ for some } a \in A\}.$$

These are subgroups of A and of B respectively.

We call f *monic* or a *monomorphism* or *injective* if $\operatorname{Ker} f = 0$.

We call f *epic* or an *epimorphism* or *surjective* if $\operatorname{Im} f = B$.

We call f an *isomorphism* if both $\operatorname{Ker} f = 0$ and $\operatorname{Im} f = B$.

An isomorphism is bijective and conversely any bijective homomorphism is an isomorphism. If f is an isomorphism then so is its inverse f^{-1}. Groups A and B are called *isomorphic*, denoted $A \cong B$, if there exists an isomorphism $f : A \to B$. The relation $\cong$ is an equivalence relation. Note that, in accordance with A.3, we write $A = 0$ rather than $A \cong 0$.

Example: any cyclic group is isomorphic either to $\mathbb{Z}_k$ for some $k \geqslant 1$ or to $\mathbb{Z}$ (see A.6).

Finitely generated (f.g.) and free abelian groups

A.9. Finitely generated abelian group Suppose that $a_1, \ldots, a_n$ are elements of A with the property that every $a \in A$ can be expressed in the form

$$a = \lambda_1 a_1 + \lambda_2 a_2 + \cdots + \lambda_n a_n$$

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with $\lambda_1, \ldots, \lambda_n$ integers (compare A.4). Then A is called *finitely generated (f.g.)* with *generators* $a_1, \ldots, a_n$. The expression for a is called an *integral linear combination* of $a_1, \ldots, a_n$.

A.10. Free f.g. abelian group Suppose that A is f.g. with generators $a_1, \ldots, a_n$. Suppose in addition that, for all integers $\lambda_1, \ldots, \lambda_n$,

$$\lambda_1 a_1 + \cdots + \lambda_n a_n = 0 \text{ implies that } \lambda_1 = \cdots = \lambda_n = 0.$$

Then A is called a *free f.g. abelian group* and $a_1, \ldots, a_n$ are called *free generators* or a *basis* for A. (Since all groups considered in the text are f.g. we usually refer there to *free abelian groups*, omitting 'f.g.' from the description.) By convention 0 is free with empty basis. It is proved in A.30 that any two bases of the same free f.g. abelian group contain the same number of elements.

Thus A is free and f.g. with basis $a_1, \ldots, a_n$ if and only if every $a \in A$ can be expressed uniquely as an integral linear combination of $a_1, \ldots, a_n$. The free f.g abelian groups are analogous to finite dimensional vector spaces, with the field over which the vector spaces are defined being replaced by the group $\mathbb{Z}$ of integers under addition. Some results about free f.g. abelian groups are in fact derived from the corresponding results about vector spaces at the end of this appendix.

Examples: (1) $\mathbb{Z}$ is f.g.; for generators we can take, for example, the two elements 2, 3, or alternatively just 1, or alternatively -1. The generator 1 is a basis; so is -1; but the two elements $2, 3$ are not.

(2) $\mathbb{Z}_k$ is f.g. but not free, since for any $a \in \mathbb{Z}_k$ we have $ka = 0$.

(3) Let $\mathbb{Z}^n$ be the group of n-tuples of integers $(\lambda_1, \ldots, \lambda_n)$ under the operation

$$(\lambda_1, \ldots, \lambda_n) + (\mu_1, \ldots, \mu_n) = (\lambda_1 + \mu_1, \ldots, \lambda_n + \mu_n).$$

Then $\mathbb{Z}^n$ is a free f.g. abelian group with basis

$$(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, 0, \ldots, 0, 1).$$

This is often called the *standard basis* for $\mathbb{Z}^n$.

★(4) The group $\mathbb{Q}$ of rational numbers under addition is not f.g. For let $a_1 = p_1/q_1, \ldots, a_n = p_n/q_n$ be rational numbers, with the p_i and q_i integers. Then any integral linear combination of $a_1, \ldots, a_n$, when reduced to its lowest terms, must have denominator a factor of $q_1 \ldots q_n$. Since there exist rational numbers without this property, e.g. $1/2q_1 \ldots q_n$, the given rational numbers do not generate $\mathbb{Q}$. ★

A.11. Free abelian group on a set The most important example for our purposes, arising in the text when defining chain groups, is as follows. Let $S = \{x_1, \ldots, x_n\}$ where the x's are distinct. Let FS be the set of expressions

$$\sum \lambda_i x_i = \lambda_1 x_1 + \cdots + \lambda_n x_n$$

where $\lambda_1, \ldots, \lambda_n$ are integers. Two elements of FS, $\sum \lambda_i x_i$ and $\sum \mu_i x_i$ say, are defined to be equal if and only if $\lambda_i = \mu_i$ for $i = 1, \ldots, n$. Addition is defined by

$$\sum \lambda_i x_i + \sum \mu_i x_i = \sum (\lambda_i + \mu_i) x_i.$$

Then FS is an abelian group; in fact it is free and f.g. with basis $1x_1 + 0x_2 + \cdots + 0x_n$, $0x_1 + 1x_2 + 0x_3 + \cdots + 0x_n, \ldots, 0x_1 + \cdots + 0x_{n-1} + 1x_n$. If we agree to omit terms with coefficient zero and to write $1x_i$ as x_i then this basis becomes just $x_1, \ldots, x_n$. We call FS the *free abelian group on* S. (Note that FS depends, strictly speaking, on the ordering of S. This dependence can be eliminated by disregarding the order in which the terms of $\sum \lambda_i x_i$ are written.) Conventionally $F\emptyset = 0$. Notice that $FS \cong \mathbb{Z}^n$, an isomorphism being given by $\lambda_1 x_1 + \cdots + \lambda_n x_n \to (\lambda_1, \ldots, \lambda_n)$.

Proposition A.12. *Let A be free and f.g. with basis $a_1, \ldots, a_n$ and let B be an abelian group. Given elements $b_1, \ldots, b_n$ in B there is a unique homomorphism $f : A \to B$ with $fa_i = b_i$ for $i = 1, \ldots, n$.*

Proof The conditions on f imply that $f(\lambda_1 a_1 + \cdots + \lambda_n a_n) = \lambda_1 b_1 + \cdots + \lambda_n b_n$ for any integers $\lambda_1, \ldots, \lambda_n$. It is straightforward to check that, A being free with basis $a_1, \ldots, a_n$, this does define a homomorphism. $\square$

For example, the boundary homomorphism $\partial_p : C_p(K) \to C_{p-1}(K)$ (see 1.19, 4.3) is completely specified by its values on the p-simplexes of K. In this example both groups are free and f.g.

Quotient groups

The definition of homology groups (4.8) depends on the idea of quotient group. Given a subgroup A of an abelian group B the quotient group B/A is a device for 'ignoring' or 'making zero' the elements of A. In the text A is the boundary subgroup of a cycle group B. Both of these are free abelian, but the following definition does not assume this.

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A.13. Cosets and quotient groups Let A be a subgroup of B. For each $b \in B$ the subset

$$\overline{b} = b + A = \{b + a : a \in A\}$$

is called a *coset of* A *in* B. (From Chapter 4 onwards we use the standard, but rather unfortunate, notation $\{b\}$ for $b + A$. It would be too confusing to use it here.) The set of cosets of A in B is denoted B/A (pronounced 'B over A') and it is an abelian group under the operation

$$\overline{b}_1 + \overline{b}_2 = \overline{b_1 + b_2}.$$

This group is called the *quotient group* of B by A.

Apart from checking the group axioms it is necessary to check that the binary operation is well-defined, i.e. that if $\overline{b}_1 = \overline{b}_3$ and $\overline{b}_2 = \overline{b}_4$ then $\overline{b_1 + b_3} = \overline{b_2 + b_4}$. This is easily done using the following fact: $\overline{b} = \overline{b}'$ if and only if $b - b' \in A$.

Examples: (1) Let $B = \mathbb{Z}$ and $A = \{nk : n \in \mathbb{Z}\}$ where k is a fixed integer $\geqslant 1$. Thus A is the set of multiples of k. Then $f : \mathbb{Z}_k \to B/A$, defined by $fn = \overline{n}$ for $n = 0, 1, \ldots, k - 1$, is an isomorphism.

(2) $B/A = 0$ if and only if $B = A$. (In particular $0/0 = 0$.)

(3) If B is finite then $|B/A| = |B|/|A|$ since every coset has $|A|$ elements, the cosets are disjoint and there are $|B/A|$ of them. In particular $|A|$ divides $|B|$: this is Lagrange's theorem in the abelian group case.

A.14. Natural projection The *natural projection* $\pi : B \to B/A$ (where A is a subgroup of B) is defined by $\pi b = \overline{b}$ for all $b \in B$. Clearly π is an epimorphism (surjective homomorphism) with kernel A.

A.15. The homomorphism theorem *Let $f : G \to H$ be a homomorphism of abelian groups. Then $G/Kerf \cong Imf$, an isomorphism h being defined by $\overline{x} \to fx$ for all $x \in G$.*

Proof Firstly, h is well defined since

$$\overline{x} = \overline{y} \Rightarrow x - y \in \mathrm{Ker} f \Rightarrow f(x - y) = 0 \Rightarrow fx = fy$$

and a homomorphism since

$$h(\overline{x} + \overline{y}) = h(\overline{x + y}) = f(x + y) = fx + fy = h\overline{x} + h\overline{y}.$$

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Finally h is monic (injective) since $fx = 0 \Rightarrow x \in \mathrm{Ker} f \Rightarrow \bar{x} = 0$ and epic (surjective) since $\mathrm{Im} f = \{fx : x \in G\}$. □

Examples: (1) Take $f : \mathbb{Z} \to \mathbb{Z}_k$ to be defined by $fn =$ residue (remainder) of n modulo k. Then f is a homomorphism with kernel the multiples of k. Compare example (1) in A.13.

★(2) Let G be the group of sequences $(\lambda_1, \lambda_2, \lambda_3, \ldots)$ where the λ_i are integers, under the operation $(\lambda_1, \lambda_2, \ldots) + (\mu_1, \mu_2, \ldots) = (\lambda_1 + \mu_1, \lambda_2 + \mu_2, \ldots)$. (Compare $\mathbb{Z}^n$ in example (3) of A.10.) Then G is an abelian group, not f.g. (in fact uncountable). Define $f : G \to G$ by $f(\lambda_1, \lambda_2, \lambda_3, \ldots) = (\lambda_2, \lambda_3, \ldots)$. Then f is epic, with kernel K isomorphic to $\mathbb{Z}$. Hence by the homomorphism theorem $G/K \cong G$; a nontrivial subgroup can be factored out without altering G up to isomorphism. It can be shown using A.27 below that that if A is f.g. and $A/B \cong A$ then $B = 0$ (see A.34(5)).★

Exact sequences

A.16. Exactness The 'sequence'

$$\cdots \to A \xrightarrow{f} B \xrightarrow{g} C \to \cdots$$

of abelian groups and homomorphisms is called *exact at B* if $\mathrm{Im} f = \mathrm{Ker}\, g$. (Note that this implies $g \circ f = 0$.) A sequence is called *exact* if it is exact at each group which is not at either end of the sequence. Exactness at an end group has no meaning. A *short exact sequence* is an exact sequence of the form

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

where, of course, the first and last homomorphisms are trivial.

The word 'exact' was first used in this sense in the book of Eilenberg & Steenrod listed in the References. There is some connexion with the concept of exact differential.

Examples: (1) $0 \to A \to 0$ is exact (i.e. exact at A) if and only if $A = 0$.

(2) $0 \to A \xrightarrow{f} B$ is exact if and only if f is monic (injective).

(3) $A \xrightarrow{f} B \to 0$ is exact if and only if f is epic (surjective).

(4) $0 \to A \xrightarrow{f} B \to 0$ is exact (at A and at B) if and only if f is an isomorphism.

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(5) For any homomorphism $f : A \to B$, there is a short exact sequence

$$0 \to \mathrm{Ker} f \xrightarrow{i} A \xrightarrow{f'} \mathrm{Im} f \to 0.$$

Here i is the inclusion ($ix = x$ for all $x \in \mathrm{Ker} f$), so that f' is the same as f except for the target, and $f'a = fa$ for all $a \in A$.

(6) If A is a subgroup of B then

$$0 \to A \xrightarrow{i} B \xrightarrow{\pi} B/A \to 0$$

is exact, where $i =$ inclusion and $\pi =$ natural projection (A.14). Conversely if

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

is exact, then $C \cong B/\mathrm{Im} f$ by the homomorphism theorem A.15.

(7) If $A \subset B$ and B is a subgroup of C, then A is a subgroup of C and $B/A, C/A, C/B$ are defined. There is a short exact sequence

$$0 \to B/A \xrightarrow{i} C/A \xrightarrow{p} C/B \to 0$$

where $i(b+A) = b+A$ and $p(c+A) = c+B$. By the homomorphism theorem A.15, $C/B \cong (C/A)/(B/A)$: the cancellation rule for subgroups.

(8) If

$$0 \to A \to B \to C \to 0$$

is exact and B is finite, then A and C are also finite, and $|B| = |A| \cdot |C|$. (Compare A.13 Ex(3) and use $C \cong B/A$.)

Direct sums and splitting

A.17. Direct sum Let $A_1, \ldots, A_n$ be abelian groups. The *direct sum* $A_1 \oplus \cdots \oplus A_n$ of the groups is the set of n-tuples $(a_1, \ldots, a_n)$ where $a_1 \in A_1, \ldots a_n \in A_n$, under the binary operation

$$(a_1, \ldots, a_n) + (a'_1, \ldots, a'_n) = (a_1 + a'_1, \ldots, a_n + a'_n).$$

Then $A_1 \oplus \cdots \oplus A_n$ is an abelian group.

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The most important case is $n = 2$, and there are 'associative' properties such as

$$A \oplus B \oplus C \cong (A \oplus B) \oplus C \cong A \oplus (B \oplus C)$$

which are very easy to check.

Examples: (1) The $\mathbb{Z}^n$ of A.10, Ex (3) is equal to $\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}$ (n summands).

(2) If A and B are f.g. with generators $a_1, \ldots, a_n$ and $b_1, \ldots, b_m$ respectively, then $A \oplus B$ is f.g. with generators $(a_1, 0), \ldots, (a_n, 0)$, $(0, b_1), \ldots, (0, b_m)$. If A and B are free and f.g. with the given generators as bases then $A \oplus B$ is free and f.g. with the given generators as basis. Similarly with more than two summands.

★(3) Suppose that p and q are coprime integers $\geqslant 1$; consider $(1, 1) \in \mathbb{Z}_p \oplus \mathbb{Z}_q$. Now $n(1, 1) = (n, n)$ $(n \in \mathbb{Z})$ and $(n, n) = (0, 0) \Leftrightarrow p|n$ and $q|n \Leftrightarrow pq|n$ since p and q are coprime. Therefore (1, 1) has order pq and $\mathbb{Z}_p \oplus \mathbb{Z}_q$, having order pq, is cyclic with (1, 1) as generator. (Conversely if p and q have a common factor > 1, then $\mathbb{Z}_p \oplus \mathbb{Z}_q$ is not cyclic.) ★

A.18. Internal direct sum Let A and B be subgroups of C such that $A \cap B = \{0\}$. The *internal direct sum* $A \oplus B$ of A and B is the subgroup of C given by

$$A \oplus B = \{a + b : a \in A \quad \text{and} \quad b \in B\}.$$

The clash of notation with A.17 is not serious since $(a, b) \to a + b$ defines an isomorphism between the direct sum and the internal direct sum of A and B, assuming of course $A \cap B = \{0\}$. Note that given c in the internal direct sum there exist *unique* $a \in A$ and $b \in B$ such that $c = a + b$.

Several theorems in the text are expressed by the exactness of certain sequences of homology groups. When applying these theorems, especially in Chapters 6, 7 and 9, the following the result is often used.

A.19. Splitting lemma *Suppose that*

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

is exact.

(1) *If there exists a homomorphism $h : C \to B$ such that $g \circ h$ is the identity on C, then B is the internal direct sum (A.18) of* $\operatorname{Im} f$ *and* $\operatorname{Im} h$, *and furthermore $B \cong A \oplus C$. We call h a* splitting homomorphism *and say that the short exact sequence is* split.

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(2) *If C is free and f.g., a splitting homomorphism always exists: the sequence is always split.*

Proof (1) Suppose that such an h exists. Then, for any $b \in B$, $(g \circ h \circ g)b = gb$, so that $b - (h \circ g)b \in \operatorname{Ker} g = \operatorname{Im} f$. Hence b can be written as the sum of elements from $\operatorname{Im} f$ and $\operatorname{Im} h$. To see that $\operatorname{Im} f \cap \operatorname{Im} h = \{0\}$ suppose that $fa = b = hc$ ($a \in A, b \in B, c \in C$). Then

$$0 = (g \circ f)a = gb = (g \circ h)c = c$$

by exactness and the given property of h. This gives the result, $b = 0$. Since f is monic, $\operatorname{Im} f \cong A$ (for example using the homomorphism theorem A.15); likewise h is monic since $g \circ h =$ identity, so that $\operatorname{Im} h \cong C$. Hence $B = \operatorname{Im} f \oplus \operatorname{Im} h \cong A \oplus C$.

(2) Let $c_1, \ldots, c_n$ be a basis for C and let $b_i \in B$ be chosen such that $gb_i = c_i, i = 1, \ldots, n$ (g is epic). Let $c \in C$. Then there exist unique integers $\lambda_1, \ldots, \lambda_n$ such that $c = \lambda_1 c_1 + \cdots + \lambda_n c_n$; define $hc = \lambda_1 b_1 + \cdots + \lambda_n b_n \in B$. Then h is a homomorphism and $g \circ h =$ identity. □

We shall make use at the end of this appendix of the following variant of the Splitting Lemma. Suppose for a moment that A, B, C are vector spaces over the same field and that f, g are linear maps. It still makes sense to say that

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

is exact: it means that f is injective, $\operatorname{Im} f = \operatorname{Ker} g$ (both are subspaces of B) and g is surjective. Suppose that C is finite-dimensional and let $c_1, \ldots, c_n$ be a basis for C.

As in (2) *above we can construct a linear map $h : C \to B$ such that $g \circ h =$ identity, and as in* (1) *we can show that this implies $B = \operatorname{Im} f \oplus \operatorname{Im} h \cong A \oplus C$, where this is the usual direct sum of vector spaces.* (In fact the same holds even if C is infinite-dimensional but the finite-dimensional case will suffice for the applications here.)

A.20. Examples (1) The sequence

$$0 \to A \xrightarrow{f} A \oplus C \xrightarrow{g} C \to 0$$

where $fa = (a, 0)$ and $g(a, c) = c$, is split by $h : C \to A \oplus C$ defined by $hc = (0, c)$. This applies whether C is free or not.

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(2) Suppose that

$$0 \to \mathbb{Z} \xrightarrow{f} B \xrightarrow{g} \mathbb{Z}_2 \to 0$$

is exact. Let $a = f1$ and let $b \in B$ satisfy $gb = 1 \in \mathbb{Z}_2$. Then $g(2b) = 1+1 = 0$, so that $2b \in \operatorname{Ker} g = \operatorname{Im} f$ and $2b = \lambda a$ for some integer λ.

Case I: λ is even. Say $\lambda = 2n$; then $h : \mathbb{Z}_2 \to B$ given by $h0 = 0, h1 = b-na$ is a splitting homomorphism, hence $B \cong \mathbb{Z} \oplus \mathbb{Z}_2$.

Case II: λ is odd. Say $\lambda = 2n+1$; then $a = 2(b-na)$ and $b = (2n+1)(b-na)$. Hence B is cyclic, generated by $b-na$ (consider separately elements of B which go to 0 and 1 under g). But a has infinite order, so B is infinite: $B \cong \mathbb{Z}$ and f takes 1 to twice a generator of B. Note that any attempt to split the sequence will fail, for a splitting homomorphism h would have to satisfy $h1 + h1 = h(1+1) = 0$, and there is no element in $\mathbb{Z}$, besides 0, of finite order.

(3) Suppose that

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

is exact and that A and C are f.g. with generators $a_1, \dots, a_n$ and $c_1, \dots, c_m$. Let $b_i \in B$ satisfy $gb_i = c_i$, $i = 1, \dots, m$. Then $fa_1, \dots, fa_n, b_1, \dots, b_m$ generate B, which is accordingly f.g.

(Proof: let $b \in B$; then $gb = \lambda_1 c_1 + \cdots + \lambda_m c_m$ for suitable integers $\lambda_1, \dots, \lambda_m$. Hence $b - \lambda_1 b_1 - \cdots - \lambda_m b_m \in \operatorname{Ker} g = \operatorname{Im} f$. Since $\operatorname{Im} f$ is generated by $fa_1, \dots, fa_n$ this completes the proof.) Thus if the sequence is split, and A and C are known, then B is determined by the splitting lemma and generators for B are given by the above result. A more precise formulation when C is free is given in (4) below.

If A and C are both free and f.g. and the above generators are bases then B is free with basis $fa_1, \dots, fa_n, b_1, \dots, b_m$.

(4) In the notation of the first part of (3) suppose that C is free, so that the sequences splits, and that $c_1, \dots, c_m$ is a basis for C. Then I claim that there is an isomorphism $\theta : B \to A \oplus C$ for which $\theta(fa_i) = (a_i, 0)$, $i = 1, \dots, n$; $\theta b_i = (0, c_i)$, $i = 1, \dots, m$. (Proof below.) Thus $fa_1, \dots, fa_n$ generate the subgroup of B corresponding under θ to $A \oplus 0$ while $b_1, \dots, b_m$ generate the subgroup of B corresponding under θ to $0 \oplus C$. We often sum this up by:

$$B \cong A \oplus C \text{ with generators } fa_1, \dots, fa_n \text{ and } b_1, \dots, b_m.$$

Other examples of this usage occur in A.22 and A.25. (Proof of the claim: Let $b \in B$. Then there exist integers λ_i, μ_j such that $b = \sum \lambda_i fa_i + \sum \mu_j b_j$

by (3). Write $\theta b = (\sum \lambda_i a_i, \sum \mu_j c_j)$. It has to be checked that this is well-defined, and is an isomorphism. The first amounts to checking that if $b = 0$ then $\theta b = (0, 0)$. Now if $b = 0$ then $gb = 0$; since $g \circ f = 0$ this implies $\sum \mu_j c_j = 0$ and so $\mu_1 = \cdots = \mu_m = 0$ by the basis property. Since f is monic this gives $\sum \lambda_i a_i = 0$. Hence θ is well-defined. The rest of the proof is left to the reader.) The result may be false if C is not free, even if the sequence splits. E.g. in (1) above take $A = \mathbb{Z}, C = \mathbb{Z}_2$, and put $a_1 = 1$, $c_1 = 1$, $b_1 = (1, 1)$.

Presentations

Let A be a f.g. abelian group with generators $a_1, \ldots, a_n$. Let F be a free abelian group of rank n with basis $x_1, \ldots, x_n$ and let $t : F \to A$ be the epimorphism defined by $t(\lambda_1 x_1 + \cdots + \lambda_n x_n) = \lambda_1 a_1 + \cdots + \lambda_n a_n$ for all integers $\lambda_1, \ldots, \lambda_n$. (Note that this t is well-defined, since any element of F is uniquely expressible as an integer linear combination of $x_1, \ldots, x_n$.) The short exact sequence

$$0 \to \operatorname{Ker} t \xrightarrow{i} F \xrightarrow{t} A \to 0$$

(i being the inclusion) is sometimes called a 'presentation' of A; however we shall use this word in a different (but closely related) sense below. Note that by the homomorphism theorem we have $A \cong F/\operatorname{Ker} t$. Note also that there is usually a wide choice available for $a_1, \ldots, a_n$ and in particular for n.

The homology groups studied in the text do in fact arise in precisely this way, as the quotient of a free abelian group by a subgroup. It is worth looking more closely at what this means. Each element of the free abelian group F with basis $x_1, \ldots, x_n$ can be expressed uniquely as an integer linear combination of $x_1 \ldots, x_n$. Given a subgroup K of F, it follows from a later result (A.33) that K is free and f.g.; however in practice we are just given generators for K, so let $y_1, \ldots, y_m$ be any generators, not necessarily free, for K. Any element of F/K can be expressed as $\lambda_1 \bar{x}_1 + \cdots + \lambda_n \bar{x}_n$ for integers $\lambda_1, \ldots, \lambda_n$, but the expression is not unique (unless $K = 0$): in fact

$$\sum \lambda_i \bar{x}_i = \sum \mu_i \bar{x}_i \Leftrightarrow \sum (\lambda_i - \mu_i)\bar{x}_i = 0 \Leftrightarrow \sum (\lambda_i - \mu_i) x_i \in K.$$

Now $x_i, \ldots, x_n$ is a basis for F so there are uniquely determined integers α_{ij} $(i = 1, \ldots, m; j = 1, \ldots, n)$ such that

$$y_i = \alpha_{i1} x_1 + \cdots + \alpha_{in} x_n \quad (i = 1, \ldots, m)$$

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and an element of F belongs to K if and only if it is an integer linear combination of the y's. The group F/K is then completely specified by the information that

$$\bar{x}_1, \ldots, \bar{x}_n \quad \text{generate it}$$

and that

$$\alpha_{i1}\bar{x}_1 + \cdots + \alpha_{in}\bar{x}_n = 0 \text{ for } i = 1, \ldots, m;$$

for the equations tell us precisely when two integer linear combinations of the $\bar{x}$'s give the same element of F/K.

The equations are called *relations* and the above description of F/K is called a presentation (by generators and relations). In practice we drop the bars, which are a notational inconvenience, and call

A.21 $$\left(\begin{array}{l} x_1, \ldots, x_n \\ \alpha_{i1}x_1 + \cdots + \alpha_{in}x_n = 0 \quad (i = 1, \ldots, m) \end{array}\right.$$

a *presentation* of F/K.

The formulation just given is designed for situations where, as with the definition of homology groups, we are given a free abelian group F and a subgroup K and wish to identify F/K. It is also possible to start with a f.g. abelian group A, choose F and t as at the beginning of the section, and let $K = \operatorname{Ker} t$. Then replacing each x_i by $tx_i = a_i \in A$ in A.21 we obtain what can be called a presentation of A. For example let $A = \mathbb{Z} \oplus \mathbb{Z}_2$, $a_1 = (1, 0)$, $a_2 = (0, 1)$. Then

$$\left(\begin{array}{ll} a_1, & a_2 \\ & 2a_2 = 0 \end{array}\right.$$

is in this sense a presentation of $\mathbb{Z} \oplus \mathbb{Z}_2$. (Compare A.28(1).) We shall stick to the former interpretation in what follows.

A.22. Example Consider the abelian group F/K with presentation

$$\left(\begin{array}{l} x, y \\ \alpha x + \beta y = 0 \end{array}\right.$$

where α and β are integers, not both zero. Let h be the greatest common divisor of α and β (the greatest common divisor of 0 and an integer k is, naturally, defined to be $|k|$), and let $\alpha = \alpha_1 h, \beta = \beta_1 h$ so that α_1 and β_1 are coprime. Hence there exist integers γ, δ such that $\alpha_1\gamma + \beta_1\delta = 1$; write

$$\alpha_1 x + \beta_1 y = u \quad \text{and} \quad -\delta x + \gamma y = v.$$

Then $x = \gamma u - \beta_1 v$; $y = \delta u + \alpha_1 v$ so that u and v generate the free abelian group F with basis x, y and so form a basis for it. (This can be checked directly;

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the general case follows from A.30 and A.34(2).) In terms of the basis u, v the relation becomes $hu = 0$, so that

$$\begin{pmatrix} u, & v \\ hu = 0 & \end{pmatrix}$$

is a presentation of the same abelian group F/K. Now this presentation is easy to interpret, for the epimorphism

$$F \to \mathbb{Z}_h \oplus \mathbb{Z}$$

$$\text{given by} \qquad \lambda u + \mu v \to (\lambda, \mu) \quad (\lambda, \mu \in \mathbb{Z})$$

has kernel the subgroup K generated by hu. Hence

$$F/K \cong \mathbb{Z}_h \oplus \mathbb{Z}.$$

Furthermore the elements of F/K which correspond to generators (1, 0) and (0, 1) of $\mathbb{Z}_h \oplus \mathbb{Z}$ are $\overline{u}$ and $\overline{v}$. We express this by saying $F/K \cong \mathbb{Z}_h \oplus \mathbb{Z}$ with generators $\overline{u}$ and $\overline{v}$. (Compare A.20(4).)

When we are given a presentation of an abelian group as in the above example, the problem is to write the group in some recognizable form. This involves changing the generators and relations, without changing the group which they present, until they become simple enough to spot the group up to isomorphism. Here are three ways in which generators and relations presenting a group F/K can be changed.

A.23. New generators and relations for old (1) We can take a new basis for F and re-express the relations in terms of the new basis. (If F has a basis containing n elements, then any n elements of F which generate F automatically form a basis – see A.30, A.34(2).) This was done in A.22, where x and y became u and v. Neither F nor K is changed, so F/K certainly remains the same.

(2) We can change the generators for K. This amounts to replacing the relations by an equivalent set – that is, one which implies and is implied by the old set. For instance we could make a succession of changes of the following kind.

$$i\text{th relation} \to \pm i\text{th relation} + \lambda(j\text{th relation})$$

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where λ is an integer and $i \neq j$. Certainly the new relations will be equivalent to the old ones and the 'left-hand sides' will define the same K. Note that the relation $0 = 0$ can be removed if ever it occurs.

(3) Consider the presentations

$$\left(\begin{array}{l} x_1, \ldots, x_k \\ r_1 = 0, \ldots, r_m = 0 \end{array}\right. \qquad \text{and} \quad \left(\begin{array}{c} x_1, \ldots, x_k, \quad x_{k+1}, \ldots, x_n \\ r_1 = 0, \ldots, r_m = 0 \\ x_{k+1} = 0, \ldots, x_n = 0 \end{array}\right.$$

(Here the r's are of course integer linear combinations of $x_1, \ldots, x_k$.) All that has happened here is that some extra generators have been added, and put equal to zero – given with one hand and taken away with the other. Writing the groups presented as F_1/K_1 and F/K respectively, we can regard F_1 as a subgroup of F; then K_1 is a subgroup of K and the inclusion $F_1 \subset F$ gives an isomorphism $F_1/K_1 \cong F/K$.

A.24. Further changes of presentation The next change of presentation involves elements from all the above. Suppose we have a presentation of F/K with generators.

$$x_1, \ldots, x_k, \quad x_{k+1}, \ldots, x_n$$

and relations

$$\begin{array}{lll} & x_{k+1} - (\text{integer linear combination of } x_1, \ldots, x_k) & = 0 \\ (R_1) & \vdots & \vdots \\ & x_n - (\text{integer linear combination of } x_1, \ldots, x_k) & = 0 \end{array}$$

(R_2) Other relations, possibly involving all of $x_1, \ldots, x_n$.

Then (R_1) tells us that $x_{k+1}, \ldots, x_n$ are redundant generators; we proceed to eliminate them. The relations (R_2) can be expressed entirely in terms of $x_1, \ldots, x_k$ by substituting from (R_1) for $x_{k+1}, \ldots, x_n$. Let the result be (R_2'). By A.23(2) the presentation with generators $x_1, \ldots, x_n$ and relations (R_1) and (R_2') defines the same F/K. Now let $x_{k+1}', \ldots, x_n'$ denote the left-hand sides of the relations (R_1). Then $x_1, \ldots, x_k, x_{k+1}', \ldots, x_n'$ are also a basis for F so that using (1) the following presentation gives the same F/K.

$$\left(\begin{array}{l} x_1, \ldots, x_k, \quad x_{k+1}', \ldots, x_n' \\ (R_1')\ x_{k+1}' = 0, \ldots, x_n' = 0 \\ (R_2')\ \text{As } (R_2),\ \text{but with } x_{k+1}, \ldots, x_n \text{ eliminated by using } (R_1). \end{array}\right.$$

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Finally we invoke (3) to say that F_1/K_1 defined by

$$\begin{cases} x_1, \dots, x_k \\ (R_2') \end{cases}$$

is isomorphic to F/K, an isomorphism being induced by the inclusion $F_1 \subset F$.

Here is an example which is typical of the calculations in the text.

A.25. Example Consider F/K defined by twelve generators $x_1, \dots, x_{12}$ and eleven relations as follows:

$$\begin{aligned} x_1 - x_8 &= 0 & x_4 + x_5 &= 0 \\ x_2 - x_8 + x_9 &= 0 & x_5 + x_6 &= 0 \\ x_9 - x_{10} - x_{11} &= 0 & x_6 + x_7 - x_{12} &= 0 \\ x_{11} - x_{12} &= 0 & x_1 - x_7 &= 0 \\ x_2 + x_3 &= 0 & x_{10} &= 0 \\ x_3 + x_4 &= 0 \end{aligned}$$

First write the relations in the following equivalent form.

$$\begin{array}{cc} x_8 = x_1 & x_6 = x_2 \\ x_7 = x_1 & x_{10} = 0 \\ x_3 = -x_2 & x_{11} = x_9 \\ x_4 = x_2 & (\text{from } x_{11} = x_9 - x_{10} = x_9) \\ (\text{from } x_4 = -x_3 = x_2) & x_{12} = x_9 \\ x_5 = -x_2 & (\text{from } x_{12} = x_{11} = x_9) \\ & x_6 + x_7 - x_{12} = 0 \\ & x_2 - x_8 + x_9 = 0 \end{array}$$

This makes it clear that all generators except x_1, x_2 and x_9 are redundant, and invoking A.24 (the first nine relations are (R_1) and the last two are (R_2)) we are reduced to

$$\begin{cases} x_1, x_2, x_9 \\ x_2 + x_1 - x_9 = 0 \\ x_2 - x_1 + x_9 = 0 \end{cases}$$

Eliminating x_9 in the same way we obtain

$$\begin{cases} x_1, x_2 \\ x_2 - x_1 + (x_2 + x_1) = 0, \end{cases} \qquad \text{i.e.} \qquad \begin{cases} x_1, x_2 \\ 2x_2 = 0 \end{cases}$$

The group F_1/K_1 now presented is clearly $\mathbb{Z} \oplus \mathbb{Z}_2$; there is an exact sequence

$$0 \to K_1 \xrightarrow{i} F_1 \xrightarrow{p} \mathbb{Z} \oplus \mathbb{Z}_2 \to 0$$

where i is inclusion and $p(x_1) = (1,0), p(x_2) = (0,1)$. Since $F_1 \subset F$ induces $F_1/K_1 \cong F/K$ we deduce that $F/K \cong \mathbb{Z} \oplus \mathbb{Z}_2$ and that an isomorphism $F/K \to \mathbb{Z} \oplus \mathbb{Z}_2$ is given by $\bar{x}_1 \to (1,0)$, $\bar{x}_2 \to (0,1)$, bars denoting cosets with respect to K. We express this by saying '$F/K \cong \mathbb{Z} \oplus \mathbb{Z}_2$ with generators $\bar{x}_1$ and $\bar{x}_2$'. (Compare A.20(4).)

What has been said about presentations so far should enable the reader to follow the comparatively few direct calculations of homology groups from presentations in the text, e.g. 4.15, 4.29(1) and (3). It is possible to give a procedure for reducing to a 'standard form' any given presentation; this procedure is based on A.22–24 and is usually described in terms of row and column operations on the 'relation matrix' (α_{ij}) of A.21. For the kind of calculations required in this book the general method is unnecessarily cumbersome, and it is easier to operate directly on the relations as in A.25. However the general method enables one to prove general theorems. The following theorem and corollary, which are given here without proof, receive full discussion in textbooks of algebra, for example, the books by Lang and Mac Lane & Birkhoff listed in the References.

A.26. Theorem *Let F be a free f.g. abelian group and K a subgroup of F. Then F/K has a presentation of the form*

$$\begin{cases} x_1, \ldots, x_n \\ \delta_1 x_1 = 0, \ldots, \delta_k x_k = 0 \end{cases}$$

where $\delta_1, \ldots, \delta_k$ are integers > 1 $\delta_1|\delta_2, \delta_2|\delta_3, \ldots, \delta_{k-1}|\delta_k$. (Of course $k \leq n$. If there are no relations we say $k = 0$.) The integer $n - k$, and the δ_i, are determined by the isomorphisms class of F/K. □

We call $n - k$ the *rank* of F/K and the δ_i are called the *invariant factors* or *torsion coefficients* of F/K. An alternative approach to rank is described in the next section. The above presentation is sometimes called the *Smith normal form*.

A.27. Corollary: Classification theorem for f.g. abelian groups

$$F/K \cong \mathbb{Z}_{\delta_1} \oplus, \cdots \oplus \mathbb{Z}_{\delta_k} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}$$

where there are $n - k$ $\mathbb{Z}$'s, i.e. rank $F/K = n - k$. □

Note that from the discussion leading to A.21 every f.g. abelian group is isomorphic to such a quotient F/K. The corollary follows from the theorem by examining the short exact sequence

$$0 \to K \to F \xrightarrow{t} \mathbb{Z}_{\delta_1} \oplus \cdots \oplus \mathbb{Z}_{\delta_k} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z} \to 0$$

where $tx_i = (0, \ldots, 0, 1, 0, \ldots, 0)$ with the 1 in the ith position, for $i = 1, \ldots, n$.

A.28. Remarks and examples (1) It is also possible to use the method of presentations to calculate quotients B/A where B is not free. In fact if B is merely f.g. then, roughly speaking, we take a presentation for B and add extra relations to express the fact that generators for A are zero. For example take $B = \mathbb{Z} \oplus \mathbb{Z}_2$ and $A =$ subgroup generated by (3,1). Then writing $a = (1, 0), b = (0, 1)$ we have the presentation

$$\left(\begin{array}{r} a, b \\ 2b = 0 \\ 3a + b = 0 \end{array} \right.$$

for B/A. This gives $B/A \cong \mathbb{Z}_6$ with generator $a + A$. (Compare the example following A.21.)

Calculations of the kind just given can be an aid when using the homomorphism theorem A.15, for if $g : B \to C$ is a homomorphism then $\operatorname{Im} g \cong B/\operatorname{Ker} g$. An example occurs towards the end of 7.5, though it so happens that there B is free.

★(2) The classification theorem A.27 implies at once that a f.g. abelian group containing no element of finite order is free. This is not easy to prove directly – indeed it is a good part of the content of A.27. For given any abelian group A the subset T of elements of finite order forms a subgroup, called the *torsion subgroup*, of A. There is a short exact sequence

$$0 \to T \to A \to A/T \to 0.$$

Certainly A/T has no element of finite order so, assuming that A is f.g., the above result implies that A/T is free and the sequence splits; $A \cong T \oplus A/T$. This is the decomposition of A.27 without the detailed structure of T. ★

Rank of a f.g. abelian group

Some of the results below can be obtained from the classification theorem A.27, but since we do not prove that theorem a self-contained discussion is given here. Except for starred examples, what follows assumes only A.1–A.20.

Let A be (as usual) an abelian group. We shall construct a vector space $\widetilde{A}$ over the rational numbers $\mathbb{Q}$ which, roughly speaking, suppresses the elements of finite order in A. Some results needed in the text do not involve the elements of finite order, and these results can be proved readily by this means. The construction is a special case of the 'tensor product' construction (see for example the books of Lang and Mac Lane & Birkhoff listed in the references) but in the present case the details are much more straightforward than in the general case, and in fact follow precisely the standard construction of the rational numbers from the integers. In tensor notation, we are constructing $A \otimes \mathbb{Q}$.

The elements of $\widetilde{A}$ are symbols ('fractions') a/n, pronounced 'a over n', where $a \in A$ and n is a non-zero integer. (The tensor notation is $a \otimes \frac{1}{n}$. Beware that in a general tensor product $A \otimes B$ not every element can be written in the form $a \otimes b$ with $a \in A$ and $b \in B$.) We identify a/n with b/m if and only if, for some non-zero integers k, ℓ, we have

$$ka = \ell b, \quad kn = \ell m.$$

(This can be expressed alternatively by saying that we impose the corresponding equivalence relation on pairs $(a, \frac{1}{n})$ and then work with the equivalence classes.) In particular $a/n = ka/kn$ for any non-zero integer k.

The rules for addition and scalar multiplication in $\widetilde{A}$ are:

$$a_1/n_1 + a_2/n_2 = (n_2a_1 + n_1a_2)/n_1n_2$$
$$r(a/n) = pa/qn \text{ where } r = p/q \text{ is a rational number and } p, q \text{ are integers.}$$

These rules respect the identifications, and $\widetilde{A}$ is a rational vector space. The zero vector is $0/1$. Note the following important fact: $a/n = 0/1$ if and only if a has finite order. (Proof: if a has finite order $k \geqslant 1$, so that $ka = 0$, then $a/n = ka/kn = 0/kn = 0/1$. If $a/n = 0/1$ then, for some non-zero integers k, ℓ we have $ka = 0$, $kn = \ell$. Hence a has finite order.) Thus passing from A to $\widetilde{A}$ suppresses the elements of finite order.

A.29. Rank Let A be f.g. with generators $a_1, \dots, a_n$. Then $a_1/1, \dots, a_n/1$ generate $\widetilde{A}$ as a rational vector space, and accordingly $\widetilde{A}$ is finite dimensional.

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The *rank* of A is defined to be the dimension of $\widetilde{A}$. (This is reconciled with A.26 and A.27 in A.32(2) below.)

A.30. Theorem *Let A be free and f.g. with basis $a_1, \ldots, a_n$. Then rank $A = n$ and every basis for A has n elements.*

Proof We show that $a_1/1, \ldots, a_n/1$ is a basis for $\widetilde{A}$, which accordingly has dimension n so that rank $A = n$. The second result follows because every basis for $\widetilde{A}$ contains n elements.

Certainly, as in A.29, the stated elements generate $\widetilde{A}$. For $a \in A$ has the form $\lambda_1 a_1 + \cdots + \lambda_n a_n$ for integers λ_i, and we have $a/m = \lambda_1/m \cdot a_1/1 + \cdots + \lambda_n/m \cdot a_n/1$ so that every element of $\widetilde{A}$ is a rational linear combination of $a_1/1, \ldots, a_n/1$. To see that these vectors are linearly independent, suppose that, for some rational numbers $r_1, \ldots, r_n$, we have $r_1(a_1/1) + \cdots + r_n(a_n/1) = 0/1$. Multiplying through by the product, k say, of the denominators of the r_i we obtain $kr_1(a_1/1) + \cdots + kr_n(a_n/1) = 0/1$, where the coefficients are now integers. Hence $kr_1 a_1 + \cdots + kr_n a_n$ has finite order and, A being free, must in fact be 0. Using $k \neq 0$ and the basis property we have $r_1 = \cdots = r_n = 0$. $\square$

Examples: $\mathbb{Z}^n$ has rank n; the trivial group has rank 0; $\mathbb{Z} \oplus \mathbb{Z}_2$ has rank 1.

Now let $f : A \to B$ be a homomorphism. Define $\widetilde{f} : \widetilde{A} \to \widetilde{B}$ by $\widetilde{f}(a/n) = fa/n$. This is well-defined (i.e. if $a/n = b/m$ then $fa/n = fb/m$) and is a linear map from $\widetilde{A}$ to $\widetilde{B}$. (What we have now defined is a 'functor' from the 'category' of abelian groups and homomorphisms to the 'category' of rational vector spaces and linear maps. Information on these concepts is in many books of abstract algebra, for example the book of Mac Lane & Birkhoff listed in the References. The next remark says that the functor is 'exact'.)

Suppose that

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact (at B). Then

$$\widetilde{A} \xrightarrow{\widetilde{f}} \widetilde{B} \xrightarrow{\widetilde{g}} \widetilde{C}$$

is exact (at $\widetilde{B}$), i.e. the subspaces $\operatorname{Im}\widetilde{f}$ and $\operatorname{Ker}\widetilde{g}$ of $\widetilde{B}$ are equal.

(Proof: $\widetilde{g}\widetilde{f}(a/n) = \widetilde{g}(fa/n) = gfa/n = 0/n = 0/1$ so $\operatorname{Im}\widetilde{f} \subset \operatorname{Ker}\widetilde{g}$. Suppose $\widetilde{g}(b/n) = 0/1$. Then gb has finite order k say, and $g(kb) = k(gb) = 0$. Hence $kb \in \operatorname{Ker} g = \operatorname{Im} f$, i.e. $kb = fa$ for some $a \in A$. Then $b/n = kb/kn = fa/kn = \widetilde{f}(a/kn)$. Hence $\operatorname{Ker}\widetilde{g} \subset \operatorname{Im}\widetilde{f}$.) Thus the operation $\sim$ takes exact sequences to exact sequences. Recall from the discussion before A.20 that every short exact sequence of vector spaces and linear maps is split.

A.31. Theorem *Suppose that*

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

is an exact sequence of f.g. abelian groups. Then

$$rank\, B = rank\, A + rank\, C.$$

Proof The corresponding short exact sequence of vector spaces splits, so $\widetilde{B} \cong \widetilde{A} \oplus \widetilde{C}$; hence

$$\operatorname{rank} B = \dim \widetilde{B} = \dim(\widetilde{A} \oplus \widetilde{C}) = \dim \widetilde{A} + \dim \widetilde{C} = \operatorname{rank} A + \operatorname{rank} C.$$

$\square$

A.32. Examples (1) If A and C are f.g. then rank $(A \oplus C) = \operatorname{rank} A + \operatorname{rank} C$. (Apply A.31 to the short exact sequence of A.20(1).)

(2) We are now in a position to see that, defining $\operatorname{rank} A = \dim \widetilde{A}$ as in A.29, the rank of the group

$$A = \mathbb{Z}_{\delta_1} \oplus \cdots \oplus \mathbb{Z}_{\delta_k} \oplus \mathbb{Z}^{n-k}$$

of A.27 really is $n - k$. For $A = T \oplus \mathbb{Z}^{n-k}$ where T is finite (i.e. $\widetilde{T} = 0$) and we have

$$\begin{aligned} \text{rank } A &= \text{rank } T + \text{rank } \mathbb{Z}^{n-k} \text{ by (1) above} \\ &= 0 + (n-k) \text{ using } \widetilde{T} = 0 \text{ and A.30.} \end{aligned}$$

(3) Let

$$0 \to A_1 \xrightarrow{f_1} A_2 \xrightarrow{f_2} \cdots \to A_{n-1} \xrightarrow{f_{n-1}} A_n \xrightarrow{f_n} 0$$

be an exact sequence of f.g. abelian groups. We can define $n - 1$ short exact sequences

$$\begin{array}{c} 0 \to \operatorname{Ker} f_i \to A_i \to \operatorname{Im} f_i \to 0 \quad (i = 1, \ldots, n-1) \\ \qquad\qquad\quad \| \\ \qquad\qquad\quad \operatorname{Ker} f_{i+1} \end{array}$$

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(Compare A.16(5).) Applying A.31 to each of these yields the *alternating sum of ranks theorem*: $\sum_{i=1}^{n}(-1)^i \operatorname{rank} A_i = 0$.

A.33. Theorem *Suppose that B is free and f.g. of rank m, and that A is a subgroup of B. Then A is free and f.g. of rank $\leqslant m$.*

Proof We proceed by induction on m. The result for $m = 1$ follows from the fact that any nontrivial subgroup of $\mathbb{Z}$ consists of the multiples of a fixed non-zero integer and is therefore itself infinite cyclic. Now assume the result is proven when B has rank $< m$ ($m \geqslant 2$) and chose a basis $b_1, \ldots, b_m$ for the free group B of rank m. Define $t : B \to \mathbb{Z}$ by $t(\lambda_1 b_1 + \cdots + \lambda_m b_m) = \lambda_m$; then $\operatorname{Ker} t$ is free of rank $m-1$, with basis $b_1, \ldots, b_{m-1}$. We shall apply the induction hypothesis to $\operatorname{Ker} t$ by letting $C = tA$ and $t' : A \to C$ be the epimorphism given by $t'a = ta$ for all $a \in A$. Then $\operatorname{Ker} t'$ is a subgroup of $\operatorname{Ker} t$ and is therefore free of rank $\leqslant m - 1$ by the induction hypothesis. The short exact sequence

$$0 \to \operatorname{Ker} t' \xrightarrow{i} A \xrightarrow{t'} C \to 0$$

is split (A.19) since C is a subgroup of $\mathbb{Z}$ and therefore free of rank $\leqslant 1$. Thus $A \cong \operatorname{Ker} t' \oplus C$ is free of rank $\leqslant m$. (Once A is known to be f.g. the inequality of ranks follows also from the fact that $\widetilde{A}$ is a subspace of $\widetilde{B}$.) □

A.34. Examples (1) Suppose that A and B are free and f.g. of the same rank and that $f : A \to B$ is an epimorphism (surjective). Then f is in fact an isomorphism. (Consider the short exact sequence $0 \to \operatorname{Ker} f \xrightarrow{i} A \xrightarrow{f} B \to 0$. This splits by A.19 and therefore $A \cong \operatorname{Ker} f \oplus B$. Hence rank $(\operatorname{Ker} f) = 0$ and since $\operatorname{Ker} f$ is free by A.33 it must be zero, i.e. f is monic (injective).) In contrast $f : \mathbb{Z} \to \mathbb{Z}$ defined by $fn = 2n$ is monic without being epic.

(2) Suppose that A is free and f.g. of rank n and that $a_1, \ldots, a_n$ generate A. Then in fact they are a basis for A. (This is just (1) applied to the epimorphism $\mathbb{Z}^n \to A$ given by $(\lambda_1, \ldots, \lambda_n) \to \lambda_1 a_1 + \cdots + \lambda_n a_n$.)

(3) Let A be free and f.g. and suppose that a and b are elements of A with the property that $\alpha a = \beta b$ for some integers α, β not both zero. Then a and b have a 'common factor', i.e. there is an element $c \in A$ of which a and b are both multiples. (When $A = \mathbb{Z}$ the hypothesis $\alpha a = \beta b$ is redundant since we can take $\beta = a$, $\alpha = b$, unless $a = b = 0$ when we can take $\alpha = \beta = 1$.) To see this suppose $a \neq 0$, so that $\beta \neq 0$, and let C be the subgroup of A generated by a and b. Denoting by $\langle a \rangle$ the infinite cyclic subgroup of C generated by a, it follows that $C/\langle a \rangle$ is finite, since a non-zero multiple βb of b belongs to $\langle a \rangle$. Hence rank $C = 1$ by A.31, i.e. C is infinite cyclic and a suitable c is a generator for C.

(4) A non-zero element a of an abelian group A is called *indivisible* if it is not a multiple of any element of A besides $\pm a$ (i.e. if $a = \lambda b$, $\lambda \in \mathbb{Z}$, $b \in A \Rightarrow \lambda = \pm 1$). Thus for example the additive group of rational numbers $\mathbb{Q}$ contains no indivisible element, and the only indivisible elements of $\mathbb{Z}$ are ± 1. Let A be a free f.g. abelian group, and let $a \in A$ be non-zero. Then *the following four statements are equivalent.*

(i) *a is indivisible.*

(ii) *$A/\langle a\rangle$ is free abelian, $\langle a\rangle$ being the (infinite cyclic) subgroup of A generated by a.*

(iii) *a is an element of some basis for A.*

(iv) *Given any basis $x_1, \ldots, x_n$ for A the unique expression $a = \lambda_1 x_1 + \cdots + \lambda_n x_n$ $(\lambda_1, \ldots, \lambda_n \in \mathbb{Z})$ has the greatest common divisor of $\lambda_1, \ldots, \lambda_n$ equal to* 1.

(The proofs (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (i) are fairly straightforward. Here are some hints. (i) $\Rightarrow$ (ii): show $A/\langle a\rangle$ has no non-zero element of finite order, perhaps using (3) above. (ii) $\Rightarrow$ (iii): Use the short exact sequence $0 \to \langle a\rangle \to A \to A/\langle a\rangle \to 0$. (iii) $\Rightarrow$ (iv): Let $a_1 = a, a_2, \ldots, a_n$ be a basis containing a. Let Λ be the $n \times n$ integral matrix expressing the a's in terms of the x's; thus the first row of Λ is $\lambda_1, \ldots, \lambda_n$. Since the a's are a basis we can consider the matrix Λ^{-1} expressing the x's in terms of the a's, and we have $\Lambda\Lambda^{-1} = I_n$, the $n \times n$ identity matrix. Hence $\det \Lambda \cdot \det \Lambda^{-1} = 1$, so $\det \Lambda = \pm 1$ (i.e. Λ is unimodular). Expanding $\det \Lambda$ by the first row shows that the greatest common divisor of $\lambda_1, \ldots, \lambda_n$ is 1. (iv) $\Rightarrow$ (i): follows from the definition of basis.)

★(5) Let $f : A_1 \to A_2$ be a homomorphism of abelian groups and let T_i be the torsion subgroup of A_i $(i = 1, 2)$, that is the subgroup of elements of finite order. Then we have $f(T_1) \subset T_2$ so f restricts to a homomorphism $f' : T_1 \to T_2$ defined by $f't = ft$ for all $t \in T_1$. (This is another example of a functor (compare the discussion following A.30), this time from the category of abelian groups and homomorphism to itself. This functor is not exact, as can be verified by considering the exact sequence $\mathbb{Z} \to \mathbb{Z} \oplus \mathbb{Z}_2 \to \mathbb{Z}$ where the homomorphisms are $m \to (0, m \bmod 2)$ and $(n, p) \to n$. But read on.)

Suppose that A_1 is f.g. and $\operatorname{Ker} f$ is finite. Then the exactness of

$$A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3$$

implies the exactness of

$$T_1 \xrightarrow{f'} T_2 \xrightarrow{g'} T_3.$$

(Hints for proof. Clearly $g' \circ f' = 0$. Identify T_1 with $0 \oplus T_1$ and use A.27 to write $A_1 \cong F_1 \oplus T_1$ with F_1 free and f.g. For simplicity write $A_1 = F_1 \oplus T_1$. Now

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$g't_2 = 0 \Rightarrow t_2 = f(x_1, t_1)$ for some $(x_1, t_1) \in A_1$. Hence $t_2 = f(x_1, 0) + f(0, t_1)$. Deduce $f(x_1, 0)$ has finite order, k say. Hence $(kx_1, 0) \in \operatorname{Ker} f$ but has infinite order unless $x_1 = 0$.)

As an example suppose B is a subgroup of the f.g. abelian group A and suppose $A/B \cong A$. We show $B = 0$. Using the exact sequence

$$0 \to B \xrightarrow{i} A \xrightarrow{\pi} A/B \to 0$$

(compare A.16 Ex (6)), and A.31, it follows that rank $B = 0$, i.e. B is finite. The result above then shows that

$$0 \to B \to T \to T' \to 0$$

is exact where T, T' are the torsion subgroups of A and A/B respectively. But T and T' have the same number of elements, so $B = 0$. ★

★(6) Consider an exact sequence

$$A \xrightarrow{f} A \oplus B \xrightarrow{g} B.$$

(i) If A and B are finite, then f is monic and g is epic. (Construct two short exact sequences from f and g as in A.16 Ex(5) and use A.16 Ex(8).)

(ii) If A and B are f.g. then $\operatorname{Ker} f$ is finite. (Show this and $B/\operatorname{Im} g$ is finite by applying the alternating sum of ranks theorem (A.32(3)) to the exact sequence $0 \to \operatorname{Ker} f \to A \to A \oplus B \to B \to B/\operatorname{Im} g \to 0$.) Hence if A, B are f.g. then with f' as in (5) above we have $\operatorname{Ker} f = \operatorname{Ker} f' = 0$ by (5) and (i). Hence: if A and B are f.g. then f is monic. It is not true in general that g is epic.

(iii) Take $B = 0$ in (ii). We have: if A is f.g. then any epimorphism $A \to A$ is an isomorphism. (Compare A.34(1). It is not hard to construct counterexamples when A is not f.g.)

(iv) If A is f.g. and B is finite then f is monic and g is epic. ★